



Ministry of Health

Arthur Davison Children's

Hospital Pathology Laboratory

Zambia

Standard Operation Procedure Technical			ID No: ADCH_TP_PCR_004	
			Version: 1	
			Supersedes: 1	
			Effective Date: 04/Nov/19	
			Next Review Date: 04/Nov/21	
TITLE: HIV-1 RNA EXTRACTION FOR DRUG RESISTANCE TESTING (GENOTYPING)				
	NAME	TITLE	SIGNATURE	DATE
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REVIEWED BY	TIONI BANDA	LABORATORY TECHNOLOGIST		
REVIEWED BY	SHADERICK MUMBA	QUALITY OFFICER		
APPROVAL AUTHORITY	ESTHER MUSONGOLE	LAB DIRECTOR		

ADCH_PL 	ARTHUR DAVISON CHILDRENS HOSPITAL PATHOLOGY LABORATORY		
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1.0. PURPOSE

To isolate and purify HIV RNA from Plasma samples in RNase/DNase free tubes using the Qiagen Viral RNA mini kit.

2.0. SCOPE

To be applied on all samples received in the lab for HIV 1 HIV Drug resistance testing.

3.0. RESPONSIBILITY

It is the responsibility of section head and trained, qualified and competent Medical Laboratory Technologist and Scientist.

4.0. PRINCIPLE

The QIAamp Viral RNA Mini Kits kit combines the selective binding properties of a silica-gel-based membrane with the speed of microspin or vacuum technology and is ideally suited for simultaneous processing of multiple samples. The sample is first lysed under highly denaturing conditions to inactivate RNases and to ensure isolation of intact viral RNA. Buffering conditions are then adjusted to provide optimum binding of the RNA to the QIAamp membrane, and the sample is loaded onto the QIAamp spin column. The RNA binds to the membrane, and contaminants are efficiently washed away in two steps using two different wash buffers. High-quality RNA is eluted in a special RNase-free buffer, ready for direct use or safe storage. The purified RNA is free of protein, nucleases, and other contaminants and inhibitors. The special QIAamp membrane guarantees extremely high recovery of pure, intact RNA in just twenty minutes without the use of phenol/chloroform extraction or alcohol precipitation.

5.0. SAMPLE REQUIREMENTS

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	Effective Date: 04/11/2019	Next Review Date: 04/11/2021	

Minimum of 1.5 ml of plasma extracted from fresh EDTA anticoagulated whole blood by centrifuging at 1000 x g for 15 minutes. Store plasma in RNAase/DNAase free plain tube.

6.0. EQUIPMENT

1. Bench Centrifuge
2. Vortex Mixer
3. 20°C and -80°C freezer
4. Timer
5. Pipettes

7.0. MATERIALS

1. Ice or ice coolers
2. Gloves
3. Micro centrifuge tubes, 1.5ml (RNAse/DNAse free).
4. Tube racks

8.0. STORAGE AND STABILITY

Sample type	Temperature	Period
EDTA Whole blood	25 to 30°C (room temperature)	2 hours
Plasma	-20 to -80°C	Up to 6 weeks
Purified RNA	-15 to -70°C	Up to 1 year

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9.0. SAFETY

1.	Follow standard laboratory practices while working with biohazardous material.
2.	Be sure to wear appropriate protective equipment, including but not limited to, safety glasses, laboratory coats and disposable gloves.
3.	If any biohazardous material is spilled, wipe it up immediately and apply disinfectant.
4.	If a sample or waste solution comes into contact with your skin, immediately apply disinfectant, and wash it off with soap and water.
5.	Do not stare into the laser transmitters beam as your eyesight maybe be severely damaged.

10.0. CALIBRATION

N/A

11.0. PROCEDURE STEPS

Preparation of reagents

1. Addition of carrier RNA to Buffer AVL

Add 310 µl Buffer AVE to the tube containing 310 µg lyophilized carrier RNA to obtain a solution of 1 µg/µl. Dissolve the carrier RNA thoroughly, divide it into conveniently sized aliquots, and store it at –30 to –15°C. Do not freeze–thaw the aliquots of carrier RNA more than 3 times. Calculate the volume of Buffer AVL–carrier RNA mix needed per batch of samples by selecting the number of samples to be simultaneously processed from Table 1.

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Table 1. Volumes of Buffer AVL and carrier RNA–Buffer AVE mix required for the QIAamp Viral RNA Mini procedure

No. samples	Vol. Buffer AVL (ml)	Vol. carrier RNA–AVE (µl)	No. samples	Vol. Buffer AVL (ml)	Vol. carrier RNA–AVE (µl)
1	0.56	5.6	13	7.28	72.8
2	1.12	11.2	14	7.84	78.4
3	1.68	16.8	15	8.4	84
4	2.24	22.4	16	8.96	89.6
5	2.8	28	17	9.52	95.2
6	3.36	33.6	18	10.08	100.8
7	3.92	39.2	19	10.64	106.4
8	4.48	44.8	20	11.2	112
9	5.04	50.4	21	11.76	117.6
10	5.6	56	22	12.32	123.2
11	6.16	61.6	23	12.88	128.8
12	6.72	67.2	24	13.44	134.4

2. Buffer AW1

Buffer AW1 is supplied as a concentrate. Before using for the first time, add the appropriate amount of ethanol (96–100%) as indicated on the bottle and in Table 2. Buffer AW1 is stable for 1 year when stored closed at room temperature (15–25°C), but only until the kit expiration date.

Table 2. Preparation of Buffer AW1

No. of preps	AW1 concentrate	Ethanol	Final volume
50	19 ml	25 ml	44 ml
250	98 ml	130 ml	228 ml

3. Buffer AW2

Buffer AW2 is supplied as a concentrate. Before using for the first time, add the appropriate amount of ethanol (96–100%) to Buffer AW2 concentrate as indicated on the bottle and in

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	Document No: ADCH_TP_PCR_004	Version: 1	
	Effective Date: 04/11/2019	Next Review Date: 04/11/2021	

Table 3. Buffer AW2 is stable for 1 year when stored closed at room temperature (15–25°C), but only until the kit expiration date.

Table 3. Preparation of Buffer AW2

No. of preps	AW2 concentrate	Ethanol	Final volume
50	13 ml	30 ml	43 ml
250	66 ml	160 ml	226 ml

Purification of viral RNA

- Record the samples you intend to extract on the RNA Extraction Form (ADCH_F149). Include one sample from the previous run that got sequenced as positive control and RNase free water for negative control
- Put orientation marks on the microfuge tube for samples with viral load less than 5000 copies/mL. Mind not to mark the lid tops as these can change position during spin. **NB:** For samples with viral load above 5000 copies/mL proceed to step 9
- Add 500ul of sample in each tube and RNase free water in the Negative control
- Put the tube in the centrifuge with orientation marks facing upwards
- Spin in a refrigerated centrifuge at 20,000-22000 x g for one hour at 4°C.
- After centrifugation stops, quickly pipette out 360ul of plasma. This should be done from the position opposite to the orientation marks. Ensure that you do it gently but within two minutes for all samples.
- Vortex the remaining 140ul.
- Quick spin for 2-4 seconds.
- Pipet 560 µl of prepared Buffer AVL containing carrier RNA into a 1.5 ml microcentrifuge tube.
- Add 140 µl plasma as well as concentrated plasma from step 1 -8 to the Buffer AVL– carrier RNA in the microcentrifuge tube. Mix by pulse-vortexing for 15 s.
- Incubate at room temperature (15–25°C) for 10 min.

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12. Briefly centrifuge the tube to remove drops from the inside of the lid.
13. Add 560 µl of ethanol (96–100%) to the sample, and mix by pulse-vortexing for 15 s. After mixing, briefly centrifuge the tube to remove drops from inside the lid.

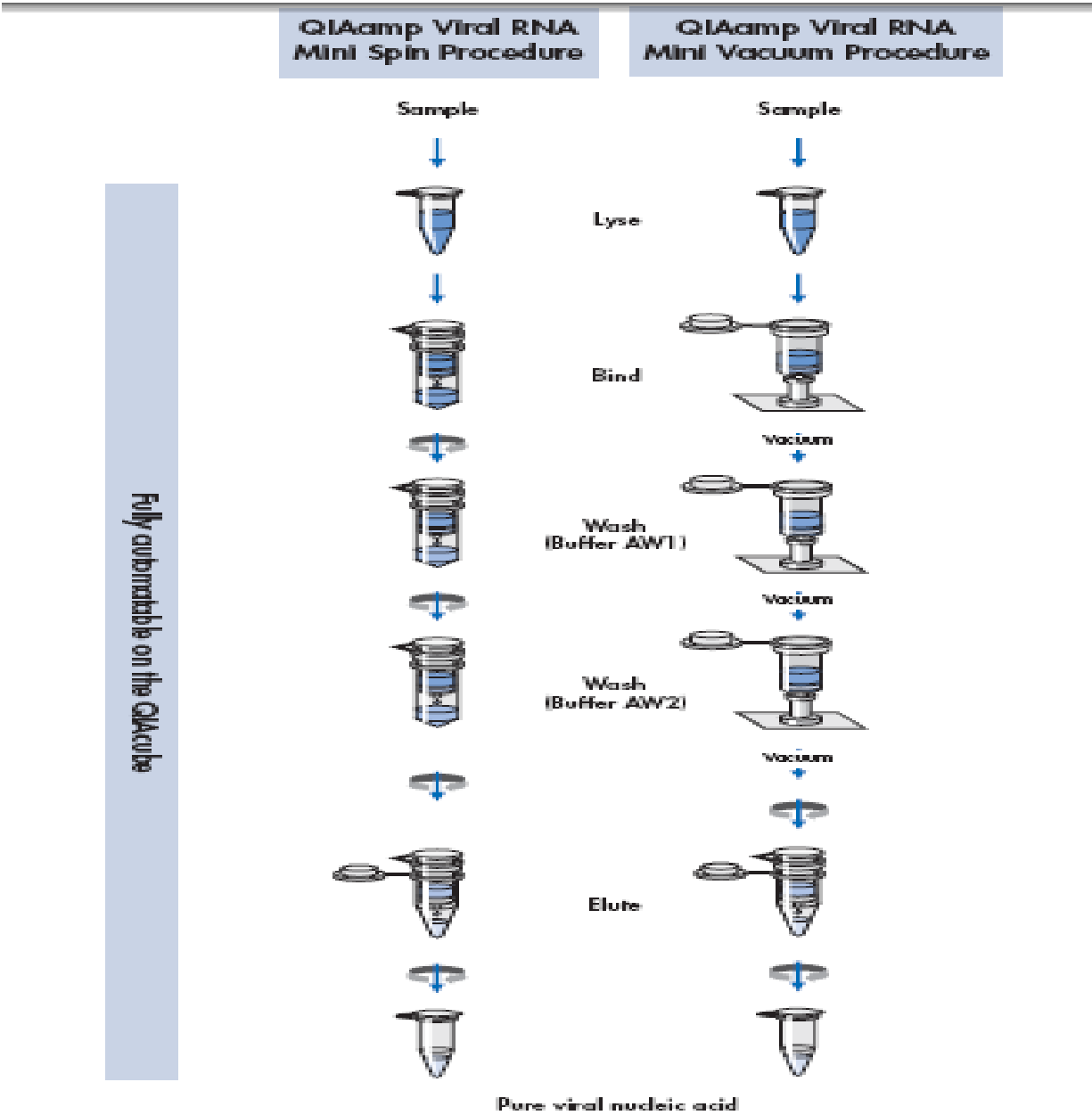
Wash and Dry

1. Carefully apply 630 µl of the solution from step 5 to the QIAamp Mini column (in a 2 ml collection tube) without wetting the rim. Close the cap, and centrifuge at 6000 x g (8000 rpm) for 1 min. Place the QIAamp Mini column into a clean 2 ml collection tube, and discard the tube containing the filtrate
2. Carefully open the QIAamp Mini column, and repeat step 11.13.
3. Carefully open the QIAamp Mini column, and add 500µl of Buffer AW1. Close the cap, and centrifuge at 6000 x g (8000 rpm) for 1 min. Place the QIAamp Mini column in a clean 2 ml collection tube (provided), and discard the tube containing the filtrate.
4. Carefully open the QIAamp Mini column, and add 500µl of Buffer AW2. Close the cap and centrifuge at full speed (20,000 x g; 14,000 rpm) for 3 min. Continue directly with step 12.17, or to eliminate any chance of possible Buffer AW2 carryover, perform step 12.16, and then continue with step 12.17
5. Recommended: Place the QIAamp Mini column in a new 2 ml collection tube (not provided), and discard the old collection tube with the filtrate. Centrifuge at full speed for 1 min.

Elution

1. Place the QIAamp Mini column in a clean 1.5 ml microcentrifuge tube (not provided). Discard the old collection tube containing the filtrate. Carefully open the QIAamp Mini column and add 60 µl of Buffer AVE equilibrated to room temperature. Close the cap, and incubate at room temperature for 1 min. Centrifuge at 6000 x g (8000 rpm) for 1 min
2. It is recommended that reverse transcription is setup immediately after extraction. However, if this cannot be done, store sample at -70°C

1. Note: see below diagram for summary.



12.0. REFERENCE RANGES

N/A

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13.0. INTERPRETATION OF THE RESULTS

N/A

14.0. LIMITATIONS OF THE PROCEDURE AND SOURCES OF ERROR

RNA is extremely sensitive to RNases and should always be prepared with due care. Hands and dust particles may carry bacteria and molds and are the most common *Sources* of RNase contamination.

15.0. REPORTING OF RESULTS

N/A

16.0. SUPPORTING DOCUMENTS

N/A

17.0. REFERENCES

QIAamp® Viral RNA Mini Handbook for purification of viral RNA from plasma, serum, cell-free body fluids, and cell-culture supernatants. Fourth Edition, December 2014.

18.0. RECORDS

RECORD TITLE	RECORD ID	CUSTODIAN/LOCATION
RNA Extraction Form	ADCH_F149	Section Head-Virology

19.0. AMENDMENT RECORD

NAME	DATE	SUMMARY OF CHANGES

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